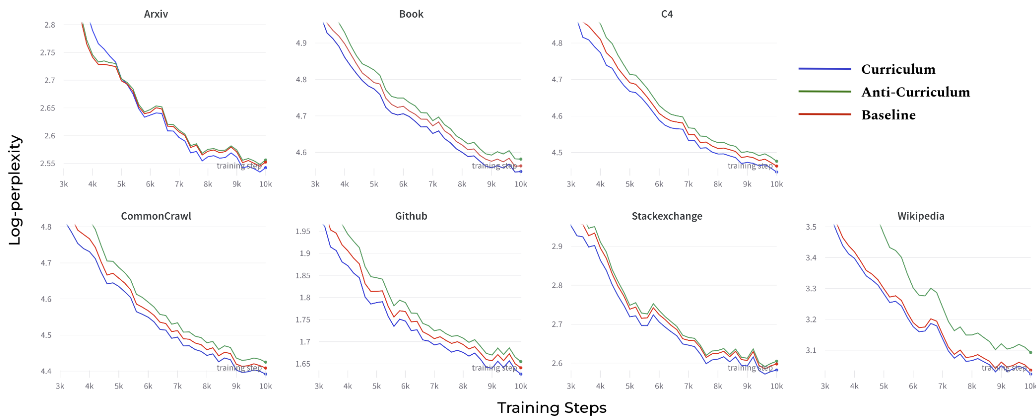

Figure 1: Domain-wise Validation Log-Perplexity on RedPajama-1B. It demonstrates IRREDUCIBLE CURRICULUM consistently outperforms BASELINE and ANTI-CURRICULUM.

IRREDUCIBLE CURRICULUM consistently improve downstream performance on all domains. We apply the irreducible curriculum across every domain within the RedPajama-1B[4] dataset, with $\lambda_0 = 50\%$, early stage checkpoint is saved at $t_0 = 2000$ steps. We train the proxy model (82M) and full model (124M) for $T = 10'000$ steps. Since the large model would learn faster than the proxy model, we set $T_c = 5000 < T$ to compensate the learning pace. According to Figure 1, IRREDUCIBLE CURRICULUM ***consistently reaches a lower validation perplexity across all 7 domains*** compared with the uniformly random baseline, even trained on a relatively small dataset. Conversely, the ANTI-CURRICULUM approach leads to a deterioration in the evaluation performance across all domains.

We also perform experiments by varying the values of λ_0 and T_c (Table 1). Notably, all the settings applying IRREDUCIBLE CURRICULUM outperform the random uniform sampling baseline, except for the case of ($\lambda_0 = 25\%, T_c = 10'000$). A possible explanation for this is that the step function determined by the proxy model inadequately addresses the differing learning paces between the large and small proxy models. On the contrary, all experiment settings in the ANTI-CURRICULUM group yield worse validation perplexity results.

Additionally, the empirical results shows that both IRREDUCIBLE CURRICULUM and ANTI-CURRICULUM approaches outperform the baseline on the 5-shot inference accuracy on MMLU benchmark [Hendrycks et al., 2021a]. Specifically, ($\lambda_0 = 25\%, T_c = 5000$) works best among all the CURRICULUM settings. Besides, it is worth noting that both CURRICULUM and ANTI-CURRICULUM improve the reasoning capability compared to baseline, which indicates the subset which contributes the most in perplexity not necessarily contains valuable factual knowledge. How to tackle with this mis-alignment towards our final target could also be a future challenge.

IRREDUCIBLE CURRICULUM reduces sharpness during training. Several recent research shows that applying sharpness aware minimization could effectively improve the generalization ability of the network [Foret et al., 2021]. Thus, we resort to inspect into how the IRREDUCIBLE CURRICULUM could impact the loss landscape by estimating its sharpness along the training trajectory. Following [Wu, 2018, Li et al., 2022b], we adopt the popular definition of sharpness as the largest eigenvalue of the Hessian matrix. We compute the Hessian of all the dense layers and record its top-eigenvalue every 200 steps along the training process. More details on sharpness computation are provided in Appendix A.

According to Fig. 2, training with IRREDUCIBLE CURRICULUM results to a lower sharpness compared with random baseline. On the opposite, the anti-curriculum would lead to a sharper training trajectory. However, whether and how this lower-sharpness correlates with the generalization ability of the network need further explorations on larger scale models and datasets.

[4]https://huggingface.co/datasets/togethercomputer/RedPajama-Data-1T-Sample

Table 1: The table presents results for different methods evaluated on average 5-shot accuracy on MMLU benchmark and average validation log-perplexity across 7 domains in RedPajama-1B. The best results are in **Red** and all the results outperform the baseline are in **Bold**. GPT-3 results refer to Hendrycks et al. [2021b].

METHOD	5-shot MMLU Acc.↑ (%)	Avg. Log Perplexity↓
RS BASELINE	22.9	3.810
GPT-3 (6.7B)	24.9	/
GPT-3 (13B)	26.0	/
GPT-3 (175B)	43.9	/
CURRICULUM ($\lambda_0 = 25\%, T_c = 2000$)	**26.0**	**3.799**
CURRICULUM ($\lambda_0 = 25\%, T_c = 5000$)	**26.9**	**3.802**
CURRICULUM ($\lambda_0 = 25\%, T_c = 10000$)	**23.5**	3.840
CURRICULUM ($\lambda_0 = 50\%, T_c = 2000$)	**23.2**	**3.800**
CURRICULUM ($\lambda_0 = 50\%, T_c = 5000$)	**24.7**	**3.794**
CURRICULUM ($\lambda_0 = 50\%, T_c = 10000$)	**24.7**	**3.800**
ANTI-CURRICULUM ($\lambda_0 = 25\%, T_c = 2000$)	**26.9**	3.819
ANTI-CURRICULUM ($\lambda_0 = 25\%, T_c = 5000$)	24.7	3.838
ANTI-CURRICULUM ($\lambda_0 = 25\%, T_c = 10000$)	**26.9**	3.892
ANTI-CURRICULUM ($\lambda_0 = 50\%, T_c = 2000$)	**26.9**	3.814
ANTI-CURRICULUM ($\lambda_0 = 50\%, T_c = 5000$)	**26.9**	3.827
ANTI-CURRICULUM ($\lambda_0 = 50\%, T_c = 10000$)	**24.9**	3.860

Figure 2: Network sharpness along training process (RedPajama-1B). The sharpness is measured as the largest eigenvalue of the hessian matrix.

5 Analysis and Failure Mode of Global Curriculum on the Data Mixture

Assuming that we have no access to the domain information during training, we can only apply the curriculum on globally on the entire data mixture. Instead of starting with top-λ_0 fraction of samples in each domains, we rank all samples in the global dataset and start sampling from top-scored λ_0 fraction, which could mostly come from one specific domain.

Here, we compare to the anti-curriculum applied on the whole data mixture, and the baseline with uniformly sampling across all instances. The model scale, architecture, and hyperparameters are the same as in Section 4.1.

Global curriculum lowers average validation perplexity and reduces sharpness. According to Fig. 3, the blue lines denote IRREDUCIBLE CURRICULUM with various combination of λ_0 and T_c[5], while the green lines apply ANTI-CURRICULUM. Across the entire domain-mixture corpus, the

[5] The combinations are (λ_0, T_c): $(25\%, 2000)$, $(25\%, 5000)$, $(50\%, 2000)$, $(50\%, 5000)$.

proposed IRREDUCIBLE CURRICULUM could still effectively reduce the validation perplexity and sharpness compared with baseline and ANTI-CURRICULUM by a large margin.

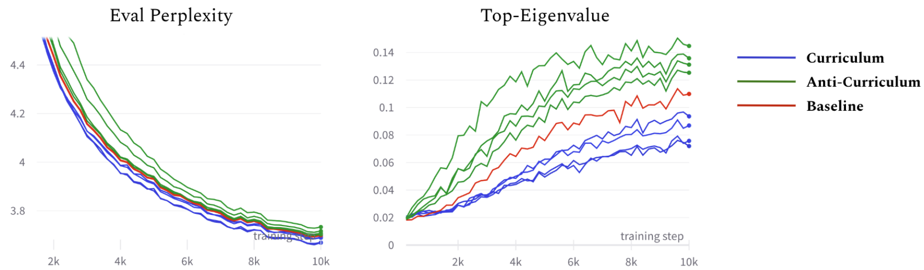

Figure 3: Validation log-perplexity on RedPajama-1B. The validation set is uniformly sampled from all the domains. The Blue lines denotes the IRREDUCIBLE CURRICULUM with different λ_0 and T_c, Green lines denotes the ANTI-CURRICULUM.

Global curriculum is not beneficial for every domain. We further evaluate the domain-wise performance with the global curriculum applied on the entire corpus. According to Fig. 5, IRREDUCIBLE CURRICULUM reduce the validation perplexity on 4 domains (C4, CommonCrawl, Book and Wikipedia), while showing the opposite effect on the other 3 domains (Arxiv, Github and Stackexchange). This can be explained by the domain-wise differences in the learnability score. Figure 4 shows that most of the globally top-ranked samples are from Wikipedia, C4 and CommonCrawl domains, which would be prioritized by the IRREDUCIBLE CURRICULUM while instances from Github and Stackexchange are mostly within the quarter with lowest scores, which are primarily learnt by the ANTI-CURRICULUM algorithm. In addition, the in-domain score distributions are highly varying among domains. It illustrates that the an intra-domain learnability (Equ. 3) ranking would offer a more fair comparison, while the naive global curriculum could risk from introducing too much inter-domain sampling bias.

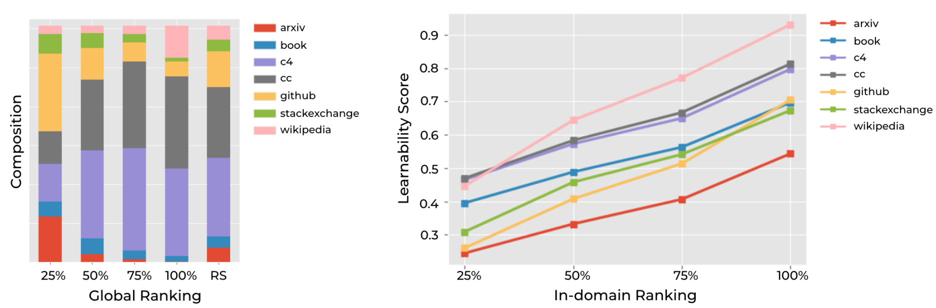

Figure 4: Learnability Scores Distribution. (Left) shows the portion of each domain in each quarter *(0-**25**%, 25%-**50**%, 50%-**75**%, 75%-**100**%)* of global ranking; (Right) shows the average learnability score of each quarter in each domain. It demonstrate that the learnability score are not necessarily comparable between different domains because of various lexical distributions and semantic knowledge.

6 Future work

Scaling up the model and training corpus. Given the emergency ability of large language models, future work is warranted to explore how this IRREDUCIBLE CURRICULUM can improve the performance of larger scale models, in particular with respect to scaling laws relating dataset size and model size.

Relation between sharpness, generalization ability and curriculum learning. In Section 4.1, we observed that the IRREDUCIBLE CURRICULUM could lead to a lower sharpness of the network along